

Supplementary Material S1: Notation Reference

This supplement provides a compact reference for all mathematical notation used in this paper. Definitions originate from Paper 1 [?] unless noted otherwise.

Cognitive State Notation (Paper 1)

Symbol	Name	Definition
σ_i	Cognitive state of agent i	$\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{I}_i, \mathcal{H}_i \rangle$
$\mathcal{B}_i$	Belief set	Probability distribution over propositions
$\mathcal{G}_i$	Goal set	Prioritized objectives with utility weights
$\mathcal{I}_i$	Intention set	Committed action plans
$\mathcal{H}_i$	History	Interaction trace (messages, actions, observations)
Σ	Set of states	$\Sigma = \{ \sigma_1, \sigma_2, \dots \}$